



PROVIDED BY MANARANJAN STUDY'S

Class 11th Physics — Chapter 1

Physical World

A complete study guide covering the fundamental concepts of physics, from the nature of physical laws to the forces that govern our universe.



Made with GAMMA

What is Physics?

Physics is the branch of science that studies the fundamental laws governing matter, energy, and their interactions in the universe.

Origin of the Word

The word *Physics* comes from the Greek word "**physis**", meaning "**nature**". In Sanskrit, it is called "**Bhautiki**" — the study of natural phenomena.

Scope of Physics

Physics deals with phenomena at all scales — from subatomic particles to galaxies. It is the most fundamental science, forming the basis of chemistry, biology, and engineering.



Branches of Physics

Physics is broadly divided into two major domains based on the scale of study:

Classical Physics

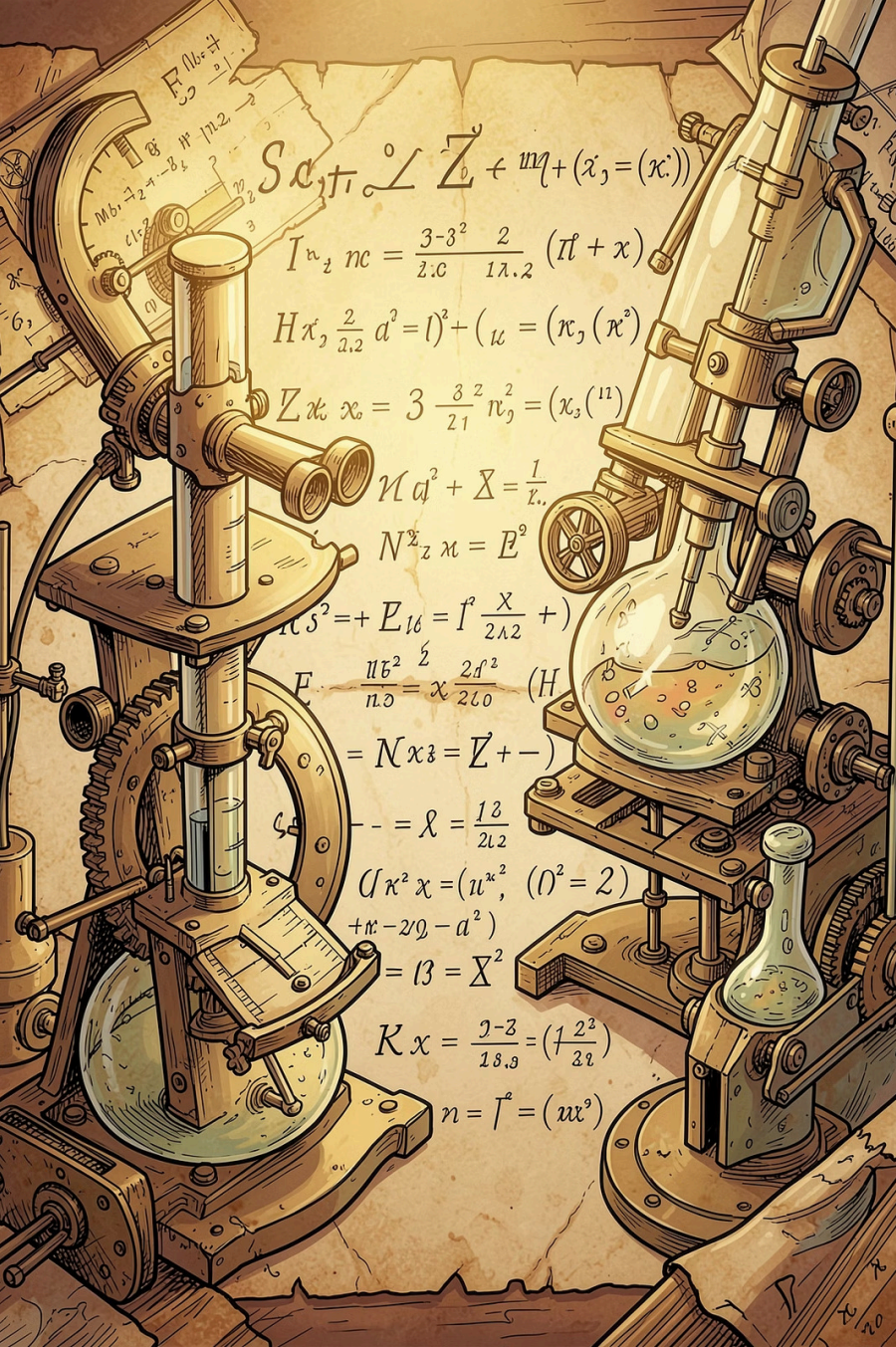
Deals with macroscopic phenomena (everyday scale)

- Mechanics — motion of objects
- Thermodynamics — heat and temperature
- Electromagnetism — electric and magnetic fields
- Optics — behavior of light

Modern Physics

Deals with microscopic phenomena (atomic and subatomic scale)

- Quantum Mechanics — behavior of particles
- Relativity — high-speed and gravitational effects
- Nuclear Physics — atomic nucleus and radioactivity
- Solid State Physics — properties of solids



Scope and Excitement of Physics

Macroscopic Domain

Laboratory, terrestrial, and astronomical scales — motion of planets, cars, projectiles, and fluids.

Microscopic Domain

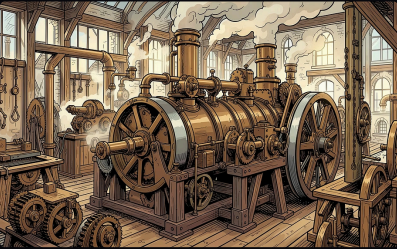
Atomic, molecular, and nuclear scales — structure of atoms, radioactivity, and quantum behavior.

Why Physics is Exciting

From discovering new particles to exploring black holes, physics continuously pushes the boundaries of human knowledge.

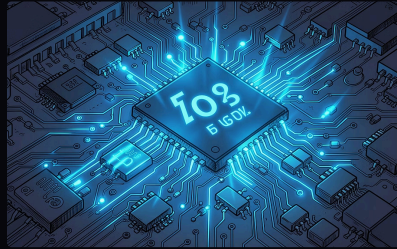
Physics, Technology & Society

Physics, technology, and society are deeply interconnected. Discoveries in physics lead to new technologies, which in turn shape society.



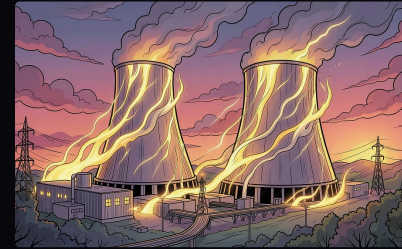
Steam Engine

Thermodynamics → Industrial Revolution → Modern transportation and power generation.



Semiconductors

Solid-state physics → Transistors → Computers, smartphones, and the digital age.



Nuclear Energy

Nuclear physics → Fission and fusion → Power plants and medical imaging (MRI, X-ray).

Fundamental Forces of Nature

There are **four fundamental forces** in nature that govern all interactions between particles and objects in the universe. These forces differ in strength, range, and the particles they affect.

Key Insight: All forces we experience in daily life — friction, tension, normal force — are ultimately derived from these four fundamental forces.



1. Gravitational Force

- weakest
- infinite range
- acts between all masses
- always attractive



2. Electromagnetic Force

- stronger than gravity
- infinite range
- acts between charged particles
- can be attractive or repulsive



3. Strong Nuclear Force

- strongest
- very short range (within nucleus)
- binds protons and neutrons



4. Weak Nuclear Force

- short range
- responsible for beta decay and nuclear reactions

Nature of Physical Laws

Physical laws are universal, consistent, and based on observation and experimentation. They describe how nature behaves under specific conditions.

1

Universality

Physical laws are the same everywhere in the universe — on Earth, in space, or in distant galaxies.

2

Conservation Laws

Certain quantities remain constant in isolated systems: **Energy, Momentum, Charge, and Mass-Energy.**

3

Based on Observation

Every physical law is derived from careful observation, experimentation, and mathematical reasoning.

4

Subject to Revision

Physical laws can be refined or replaced when new evidence emerges — this is the strength of science.

Conservation Laws in Physics



Conservation of Energy

Energy cannot be created or destroyed — only transformed from one form to another.



Conservation of Momentum

Total momentum of an isolated system remains constant if no external force acts.



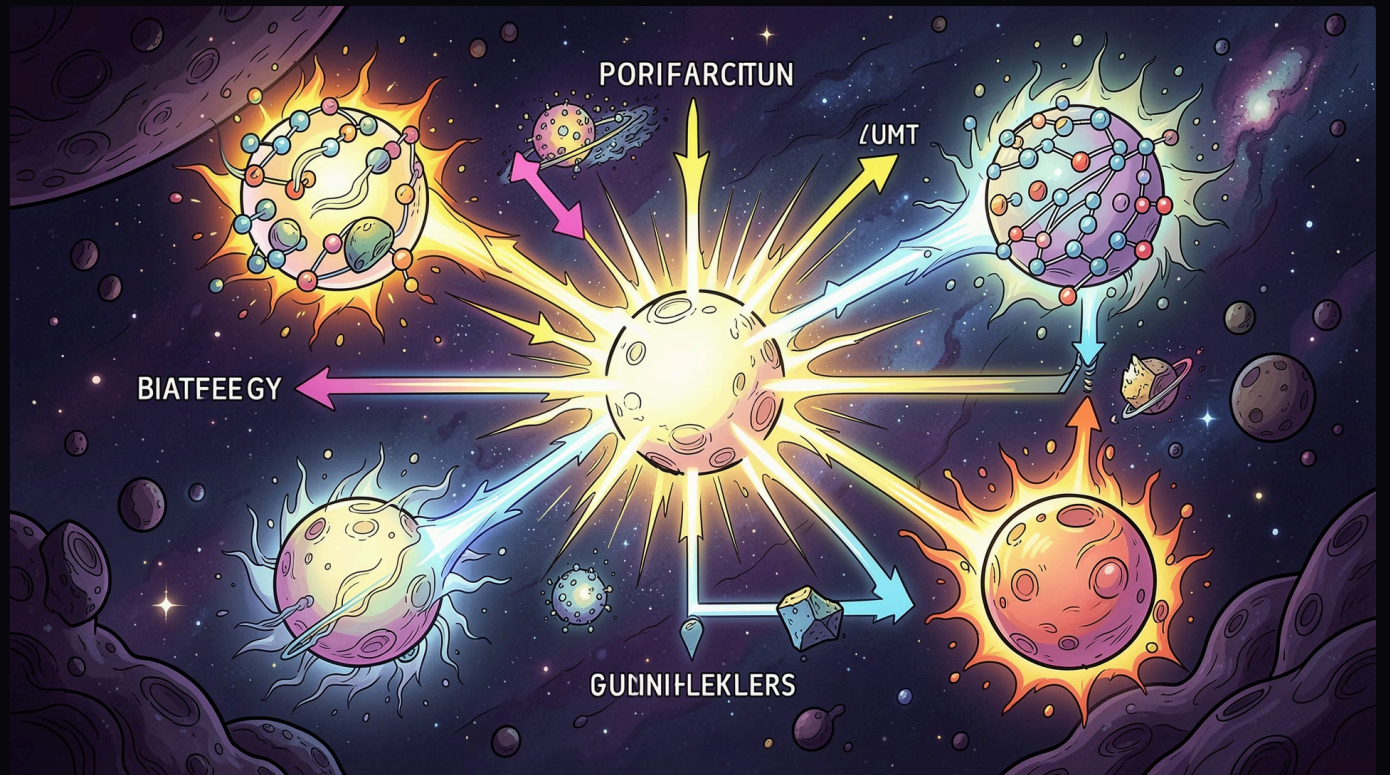
Conservation of Charge

Total electric charge in an isolated system is always conserved.



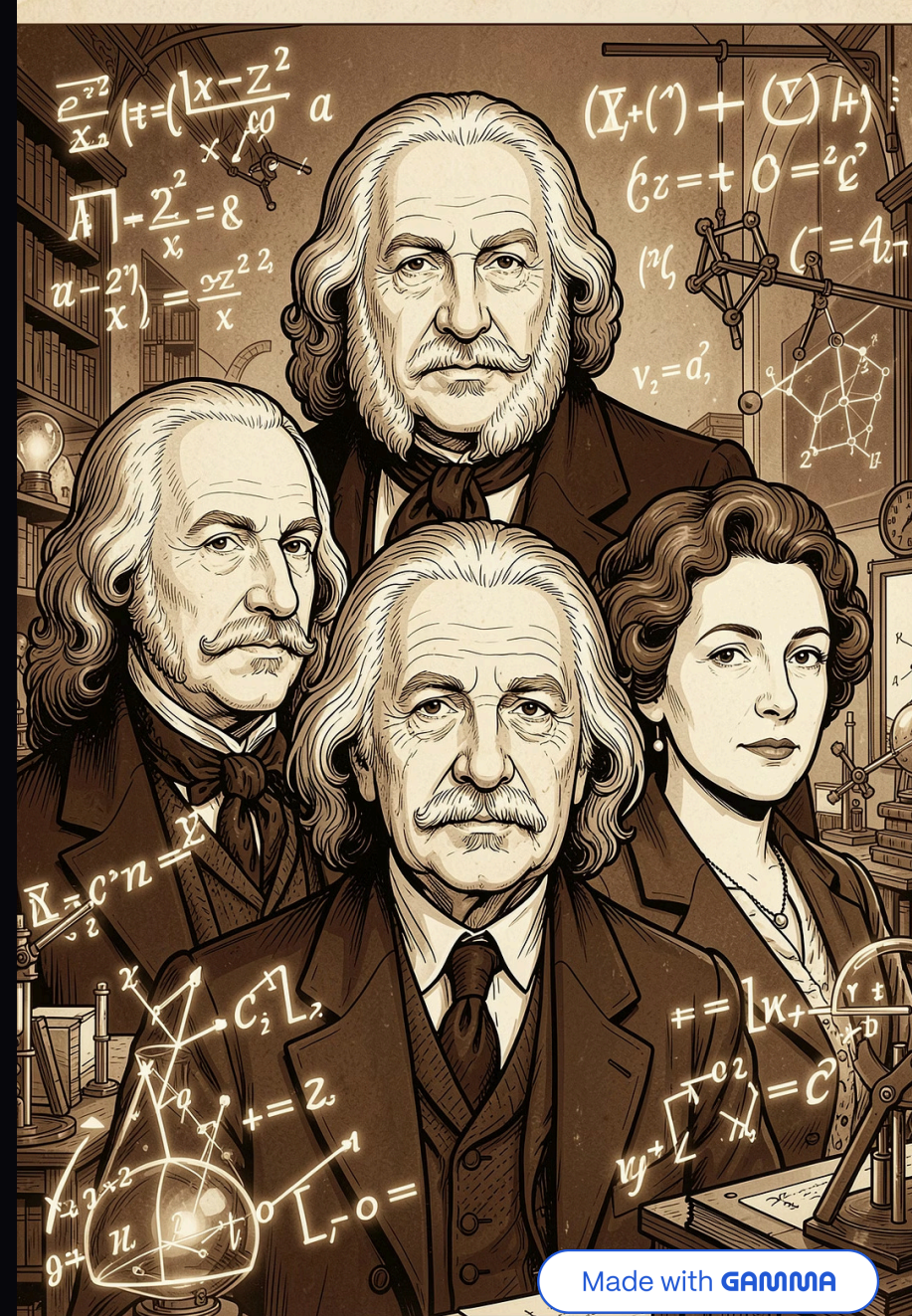
Conservation of Mass-Energy

Mass and energy are interchangeable ($E=mc^2$); their total is conserved.



Great Physicists & Their Contributions

- 1 — Isaac Newton (1643–1727)
Laws of Motion, Universal Gravitation, Calculus
- 2 — Michael Faraday (1791–1867)
Electromagnetic Induction, Laws of Electrolysis
- 3 — Albert Einstein (1879–1955)
Theory of Relativity, Photoelectric Effect, $E=mc^2$
- 4 — Niels Bohr (1885–1962)
Atomic Structure, Quantum Theory of the Atom



Chapter 1 — Quick Revision

Key takeaways from **Physical World** to remember for your exams:

→ **Physics = Study of Nature**

Physics explores the fundamental laws of matter, energy, and their interactions at all scales.

→ **Two Domains**

Classical (macroscopic) and **Modern** (microscopic) physics together cover the full scope of the universe.

→ **Four Fundamental Forces**

Gravitational, Electromagnetic, Strong Nuclear, and Weak Nuclear forces govern all interactions.

→ **Conservation Laws**

Energy, Momentum, Charge, and Mass-Energy are always conserved in isolated systems.



Provided by Manaranjan Study's — Master Physics, One Chapter at a Time!

By manaranjan sahu